

Xi Shi

Ph.D. Student in Computer Science, University of Central Florida

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Research: multi-agent LLM systems · LLM inference efficiency (KV-cache reuse) · post-training · AI safety

Education

University of Central Florida – Ph.D. in Computer Science (Advisor: [Prof. Qian Lou](#)) Aug 2025 – May 2029 (expected)

Texas A&M University – M.S. in Computer Science (Advisor: [Prof. Ruihong Huang](#)) Aug 2023 – May 2025

Jilin University – B.Sc. in Computer Science and Technology Sep 2019 – Jun 2023

Publications

[C1] **Xi Shi**, Mengxin Zheng, Qian Lou. “Learning Latency-Aware Orchestration for Multi-Agent Systems.” **NeurIPS 2026** (Main Track). [arXiv:2601.10560](#) · [Code](#)

[U1] **Xi Shi**, Mengxin Zheng, Qian Lou. “KVShareArena: KV-Cache Reuse Across Contexts and Model Checkpoints.” *Under review at ICLR 2027*. [arXiv:2609.10266](#) · [Code](#)

[U2] **Xi Shi**, Qian Lou. “Learning to Route Safely in Multi-Agent LLM Systems.” *Under review*, 2026.

[C2] Md Messal Monem Miah, Adrita Anika, **Xi Shi**, Ruihong Huang. “Hidden in Plain Sight: Evaluation of the Deception Detection Capabilities of LLMs in Multimodal Settings.” **ACL 2025 (Main)**.

[C3] Kangda Wei, **Xi Shi**, Jonathan Tong, et al. “LegalCore: A Dataset for Event Coreference Resolution in Legal Documents.” **Findings of ACL 2025**.

[C4] Yin Wang, Xuyang Xu, Ziteng Gao, **Xi Shi**. “A Prior Probability of Speaker Information and Emojis Embedding Approach to Sarcasm Detection.” **IEEE TALE 2021**.

Research Experience

University of Central Florida Orlando, FL
Graduate Research Assistant, advised by Prof. Qian Lou Sep 2025 – Present

KVShareArena: Benchmarking KV-Cache Reuse for LLM Inference First author, ICLR 2027 submission [U1]

- Built an open benchmark for reusing KV caches when the preceding context or the model checkpoint changes (RAG, multi-agent pipelines), comparing **11 repair and compression methods** from 6 classes under one protocol, with a 3-call plug-in interface and a public leaderboard.
- Showed that repairs recomputing no document tokens cut **TTFT by 81–94%** while compression saves 25–72% of KV memory at lower quality; found a trained adapter degrades in 12 of 18 cross-checkpoint tests, and retraining recovers up to 0.25 F1 at ~20 GPU-hours.

LAMaS: Latency-Aware Orchestration for Multi-Agent LLM Systems First author, NeurIPS 2026 [C1]

- Led a latency-aware orchestration framework for multi-agent LLM systems that optimizes the critical execution path.
- Learned execution graphs via constrained optimization with critical-path-aware credit assignment, plus a lightweight runtime controller that prunes redundant agent calls, cutting **end-to-end latency by over 50%** on four benchmarks at competitive or better accuracy; code released.

Safety-Aware Routing for Multi-Agent LLM Systems First author, under review [U2]

- Built a constrained RL framework whose adaptive Lagrangian router jointly selects collaboration topology, agent roles, and LLM assignments, improving jailbreak resistance and robustness to malicious agents under an explicit clean-accuracy floor; also evaluated confidential-context leakage.

Coverage-Preserving Multi-Domain On-Policy Distillation Ongoing research, Apr 2026 – present

- Studying how to merge domain experts into one small model with multi-teacher on-policy distillation: a Qwen3-4B student learns from two 30B-A3B MoE teachers (science, long-document QA) with per-domain routing, built on slime (Megatron-LM + SGLang) with custom multi-teacher and reward patches.
- Raised science accuracy by **+5.6** and LongBench QA F1 by **+2.2** points (paired bootstrap, both significant) without hurting GPQA; showed that a single teacher hurts the other domain (up to **−5.2** points, **−9** on GPQA) by transferring response habits, while domain routing keeps the worst-domain drop at zero.

Texas A&M University College Station, TX
Graduate Student Researcher, advised by Prof. Ruihong Huang Feb 2024 – May 2025

- Evaluated LLMs and multimodal models on deception detection (ACL 2025 [C2]) and legal event coreference (LegalCore [C3]); studied jailbreak robustness of LLMs and VLMs.

Skills

LLM Serving: vLLM, SGLang, FlashAttention; KV-cache reuse and inference profiling (prefill, TTFT, KV memory)

ML: PyTorch, HF Transformers, slime (Megatron-LM + SGLang); on-policy distillation, GRPO, SFT, constrained RL

Languages & Tools: Python, C/C++, Bash, SQL; Git, Linux, Slurm on HPC clusters, $\text{\LaTeX}$

Service: Reviewer, ACL Rolling Review (ARR), 2026